English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Probability and Statistics, CS Year 3, 2020/2021. Name:Student no.: *Your grade is 1 + the number of points. Good luck! Work on* $(\Omega, \mathcal{F}, \mathbb{P})$. *Define* $\mathbb{N}_0 = \mathbb{N} \cup \{0\}$. *Unless otherwise stated,* $n, k, i \in \mathbb{N}$.

Problem 1. An attacker sends a phishing email to a lecturer. At a random time $X \sim U(0,1)$ during the next day, the attacker hides all traces but also loses the opportunity to benefit. Thus $f_X(x) = 1$ on $[0,1]$ and 0 elsewhere. The lecturer responds after an independent exponential time with mean two days: $Y \sim \text{Exp}(1/2)$, with $f_Y(y) = \frac{1}{2}e^{-y/2}$ for $y \geq 0$ and 0 elsewhere.

1. (0.75 points) What is the probability that the attack succeeds? Approximately what fraction of many attempts will succeed?
2. (0.25 points) What is the probability the attacker benefits given that the lecturer responds after exactly 12 hours?

Problem 2. Two faces of a fair die are white, two green and two red. Roll it twice. Let X count white results and Y red results. Find:

1. (0.5 points) the joint distribution and correlation of X, Y. Are they independent?
2. (0.25 points) $\mathbb{P}(X = 1 \mid Y = 1)$;
3. (0.25 points) $\mathbb{P}(X > Y)$.

Problem 3. Daily accident counts in Bulgaria are modelled by independent Poisson variables $X_1, X_2, \ldots$ with $\lambda = 10$, so $\mathbb{P}(X_1 = k) = e^{-\lambda}\lambda^k/k!$ for $k \in \mathbb{N}_0$. Set $S_n = X_1 + \cdots + X_n$.

1. (0.25 points) Find $\mathbb{E}S_n$ and DS_n.
2. (0.75 points) If T is the total number of serious accidents in 2021, show $\mathbb{P}(3529 \leq T \leq 3771) \approx 0.954$ and carefully justify the approximation.

Problem 4. Each of $n > 2$ people tosses a fair coin. A person wins if their result differs from everyone else's. If there is no winner, all toss again.

1. (0.25 points) Let X count rounds until a winner. Find its expectation and variance.
2. (0.75 points) After the first winner leaves, continue with the remaining $n-1$ players, then $n-2$, and so on until two remain. What is the expected number of rounds? If winner k receives $100(n-k)$, what is the most you would pay to enter?

Problem 5. A fan collects an album of $n = 500$ different Euro 2020 stickers (in 2021), buying random stickers. Let X be the number needed to collect all types.

1. (0.5 points) Let $X_1 = 1$, and for $i \geq 2$ let X_i count purchases after obtaining the $(i-1)$st distinct sticker through obtaining the ith, inclusive. For example, the sequence $\mathbf{5}, \mathbf{12}, 5, \mathbf{111}, 12, 111, 5, \mathbf{13}$ gives $X_1 = 1, X_2 = 1, X_3 = 2, X_4 = 4$. What is the distribution of X_i? Find $\mathbb{E}X$.
2. (0.5 points) Estimate the probability that at least $2n$ purchases are needed.